Math

27 QUESTIONS

DIRECTIONS

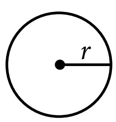
The questions in this section address a number of important math skills.
Use of a calculator is permitted for all questions.

NOTES

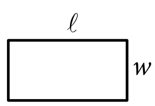
Unless otherwise indicated:

- All variables and expressions represent real numbers.
- Figures provided are drawn to scale.
- All figures lie in a plane.
- The domain of a given function f is the set of all real numbers x for which $f(x)$ is a real number.

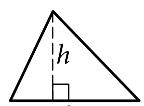
REFERENCE



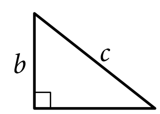
$A = \pi r^2$
$C = 2\pi r$



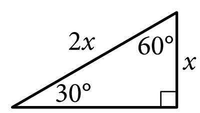
$A = \ell w$

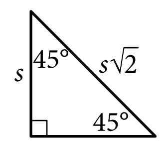


$A = \frac{1}{2}bh$

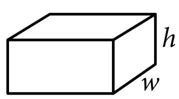


$c^2 = a^2 + b^2$

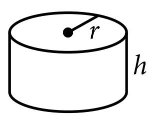




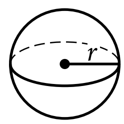
Special Right Triangles



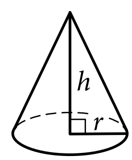
$V = \ell wh$



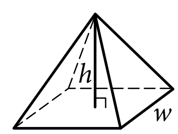
$V = \pi r^2 h$



$V = \frac{4}{3}\pi r^3$



$V = \frac{1}{3}\pi r^2 h$



$V = \frac{1}{3}\ell wh$

The number of degrees of arc in a circle is 360.

The number of radians of arc in a circle is 2π.

The sum of the measures in degrees of the angles of a triangle is 180.

CONTINUE

For multiple-choice questions, solve each problem, choose the correct answer from the choices provided, and then circle your answer in this book. Circle only one answer for each question. If you change your mind, completely erase the circle. You will not get credit for questions with more than one answer circled, or for questions with no answers circled.

For student-produced response questions, solve each problem and write your answer next to or under the question in the test book as described below.

- Once you've written your answer, circle it clearly. You will not receive credit for anything written outside the circle, or for any questions with more than one circled answer.
- If you find **more than one correct answer**, write and circle only one answer.
- Your answer can be up to 5 characters for a **positive** answer and up to 6 characters (including the negative sign) for a **negative** answer, but no more.
- If your answer is a **fraction** that is too long (over 5 characters for positive, 6 characters for negative), write the decimal equivalent.
- If your answer is a **decimal** that is too long (over 5 characters for positive, 6 characters for negative), truncate it or round at the fourth digit.
- If your answer is a **mixed number** (such as $3\frac{1}{2}$), write it as an improper fraction (7/2) or its decimal equivalent (3.5).
- Don't include **symbols** such as a percent sign, comma, or dollar sign in your circled answer.

1

Which expression is equivalent to $6x + 5x + 4y$?

A) $15x$

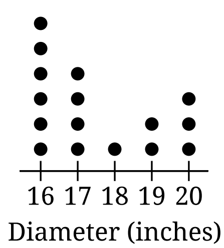
B) $15y$

C) $11x + 4y$

D) $30x + 4y$

2

To study the characteristics of sea stars in a group of tide pools, researchers measured the diameter of the sea stars within the tide pools. The dot plot gives the diameter, to the nearest inch, of each of the sea stars in these tide pools.



Based on the dot plot, how many sea stars had a diameter, to the nearest inch, of 16 inches?

A) 16

B) 6

C) 4

D) 1

3

A rectangle has a length of 56 inches and a width of 28 inches. What is the area, in square inches, of the rectangle?

A) 28

B) 84

C) 168

D) 1,568

4

$$10x = 110$$
$$6x - 63 = y$$

The solution to the given system of equations is (x, y). What is the value of y?

A) 63

B) 11

C) 10

D) 3

5

The function f is defined by $f(x) = 9(2x + 3)$. For what value of x does $f(x) = 63$?

A) 2
B) 5
C) 7
D) 30

6

$$10x = 86$$

What value of x is the solution to the given equation?

7

$$y = 3{,}600(a)^x$$

The given equation, where a is a positive constant, gives the predicted number of bacteria, y, in a growth medium x hours after the number of bacteria was initially measured. According to the equation, what was the predicted number of bacteria initially measured in the growth medium?

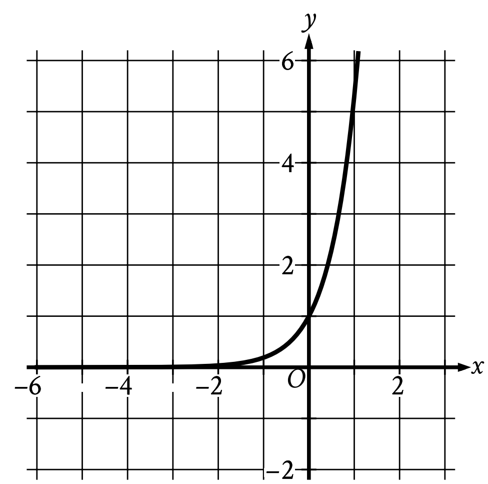
8

Leo goes to a packing store to buy containers and tape. Leo has \$15. Each container costs \$1.87 and each roll of tape costs \$2.40. Which inequality represents the relationship between the number of containers, c, and the number of rolls of tape, t, Leo can buy?

A) $1.87c + 2.40t \leq 15$
B) $1.87c + 2.40t \geq 15$
C) $2.40c + 1.87t \leq 15$
D) $2.40c + 1.87t \geq 15$

9

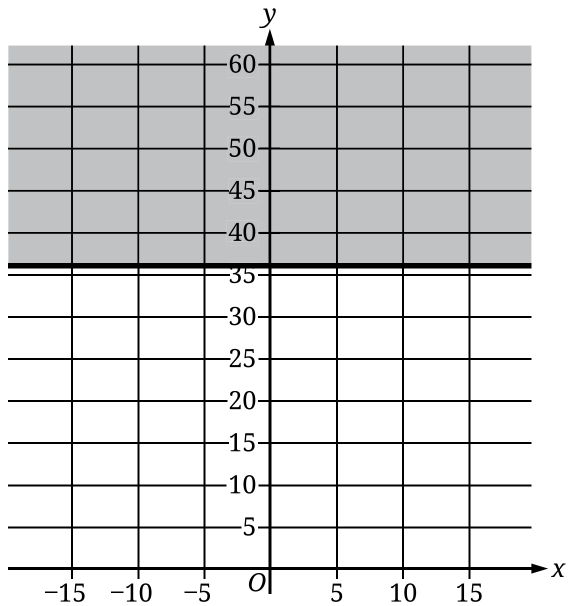
The graph of the function f is shown, where $y = f(x)$.



Which of the following best describes the function f?

A) Decreasing exponential
B) Increasing exponential
C) Decreasing linear
D) Increasing linear

10



The shaded region shown in the graph represents all the solutions to which inequality?

A) $x \leq 36$
B) $x \geq 36$
C) $y \leq 36$
D) $y \geq 36$

11

There are 240 players in a tennis competition that includes 4 rounds of matches. Each player in the competition will play a match against another player in round 1. At the end of each round, the player who loses the match is eliminated and the player who won the match advances to the next round to play a match against another winning player. Which equation gives the number of players, p, eliminated at the end of round r, where $r \leq 4$?

A) $p = 15\left(\frac{1}{2}\right)^r$

B) $p = 15(2)^r$

C) $p = 240\left(\frac{1}{2}\right)^r$

D) $p = 240(2)^r$

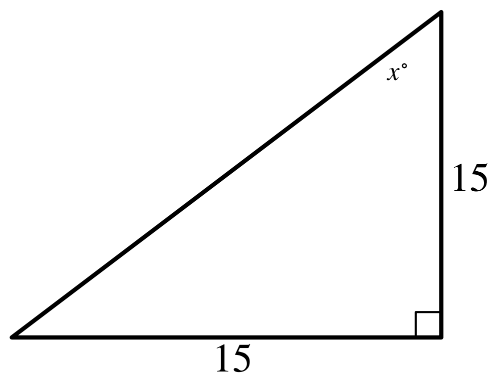
12

Line k is defined by $y = 6x + 4$. Line j is parallel to line k in the xy-plane and passes through the point $(0, 5)$. Which equation defines line j?

A) $y = 6x + 5$
B) $y = -5x + 5$
C) $y = -6x + 5$
D) $y = 5x + 5$

CONTINUE

13



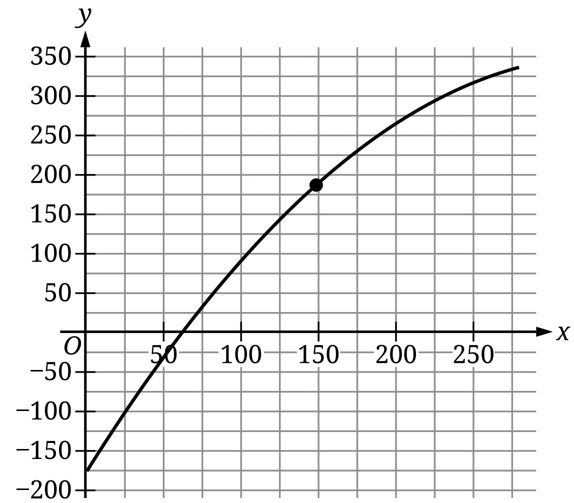
Note: Figure not drawn to scale.

In the triangle shown, what is the value of x?

14

What is the radius of the circle in the xy-plane defined by $(x+2)^2+(y+5)^2=169$?

15



The graph shows the estimated boiling point y, in degrees Celsius, of a normal paraffin with a molecular weight of x grams per mole, where $1 \leq x \leq 280$. Which statement is the best interpretation of the point $(149.02, 186.05)$?

A) A normal paraffin with a molecular weight of 186.05 grams per mole has an estimated boiling point of 149.02 degrees Celsius.

B) A normal paraffin with a molecular weight of 149.02 grams per mole has an estimated boiling point of 186.05 degrees Celsius.

C) The minimum estimated boiling point for normal paraffins corresponds to a paraffin with a molecular weight of 149.02 grams per mole and an estimated boiling point of 186.05 degrees Celsius.

D) The maximum estimated boiling point for normal paraffins corresponds to a paraffin with a molecular weight of 149.02 grams per mole and an estimated boiling point of 186.05 degrees Celsius.

16

For the polynomial function f, the graph of $y = f(x)$ in the xy-plane passes through the points $(-5, 0)$, $(1, 0)$, and $(4, 0)$. Which of the following must be a factor of $f(x)$?

A) $x + 1$
B) $x + 4$
C) $x - 1$
D) $x - 5$

17

x	$g(x)$
1	32
2	28
3	24
4	20

For the linear function g, the table shows four values of x and their corresponding values of $g(x)$. The function can be written as $g(x) = mx + b$, where m and b are constants. What is the value of b?

A) 4
B) 16
C) 32
D) 36

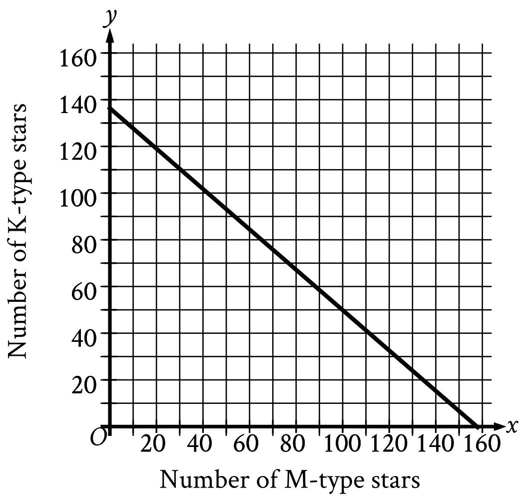
18

x	24	30	32
$f(x)$	-8	-8	8

For the quadratic function f, the table shows three values of x and their corresponding values of $f(x)$. Which equation defines f?

A) $f(x) = (x - 24)(x - 30) + 4$
B) $f(x) = (x - 24)(x - 30) - 8$
C) $f(x) = (x - 8)(x - 32) + 32$
D) $f(x) = (x - 8)(x - 32) - 32$

19



A certain open star cluster contains M-type stars and K-type stars. The estimated total mass of M-type and K-type stars in this open star cluster is 127,882 quettagrams. The graph shown models the possible combinations of the number of M-type stars, x, and K-type stars, y, that could be in this open star cluster if all the M-type stars have the same estimated mass and all the K-type stars have the same estimated mass. Based on the graph, which of the following is closest to the estimated mass, in quettagrams, of each M-type star in this cluster?

A) 811
B) 938
C) 51,904
D) 75,978

20

$$\sqrt[3]{p^2} = t^{\frac{9}{7}}$$

In the given equation, $p > 1$ and $t > 1$. If $t = p^{3n-1}$, where n is a constant, what is the value of n?

21

The number a is 55% less than the number b. The number b is 320% greater than 160. What is the value of a?

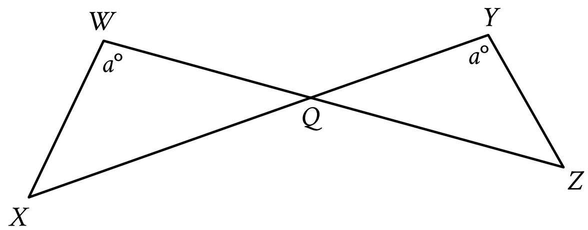
22

$$f(x) = -6x^2 + 60x - 126$$

The function f is defined by the given equation. Which of the following equivalent forms of the equation displays the maximum value of the function as a constant or coefficient?

A) $f(x) = -6x^2 + 42x + 18x - 126$
B) $f(x) = -6x(x - 7) + 18(x - 7)$
C) $f(x) = -6(x - 5)^2 + 24$
D) $f(x) = -6(x - 7)(x - 3)$

23



Note: Figure not drawn to scale.

In the figure shown, $\overline{WZ}$ and $\overline{XY}$ intersect at point Q, $YQ = 21$, $WQ = 70$, $WX = 60$, and $XQ = 120$. What is the length of $\overline{YZ}$?

A) 18
B) 36
C) 120
D) 200

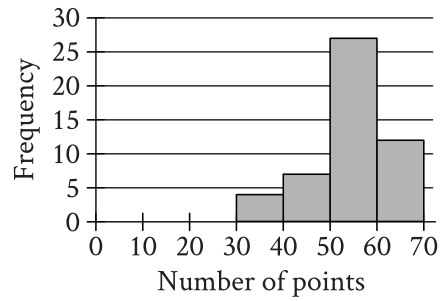
24

$$52(x^3 + 64)(x^4 - 81) = 0$$

How many distinct real solutions does the given equation have?

A) Exactly two
B) Exactly three
C) Exactly five
D) Exactly seven

25



The histogram summarizes data set A, which represents the number of points per player earned by 50 players of a game. A new player earns 18 points playing the game, and this number of points is added to data set A to create data set B with 51 values. Which of the following must be true?

I. The median number of points per player for data set B is less than the median number of points per player for data set A.

II. The mean number of points per player for data set B is less than the mean number of points per player for data set A.

A) I only
B) II only
C) I and II
D) Neither I nor II

26

In triangle XYZ, the measure of angle X is 90°. Point W lies on segment YZ, and segment WX is perpendicular to segment YZ. The length of segment WY is 572, and the length of segment WX is 429. What is the value of $\tan Z$?

A) $\frac{3}{5}$

B) $\frac{3}{4}$

C) $\frac{4}{5}$

D) $\frac{4}{3}$

27

An area of 46.00 square nautical miles is equivalent to k square kilometers. To the nearest tenth, what is the value of k? (1 nautical mile $=$ 1.852 kilometers)

STOP

If you finish before time is called, you may check your work on this module only. Do not turn to any other module in the test.